

PAPER_817: GRMHD Binary BH Merger Accretion Modulation UQFF

Unified Quantum Field Framework — Whitepaper 817

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Abstract

This paper derives the GRMHD (General Relativistic MagnetoHydroDynamic) binary black hole merger accretion modulation terms for the Quadriadic UQFF, based on arXiv:2309.03949. In equal-mass spinning binary BH systems embedded in magnetized gas, the accretion rate is modulated by the orbital frequency f_{orb} , producing characteristic lumpiness factors $A \cdot \cos(\omega t + \phi)$. The Poynting luminosity L_{Poynt} couples to spin parameter $a_\bullet$ to generate the electromagnetic counterpart. These terms enter UQFF Layers 2-4 as binary orbital modulation corrections.

1. Introduction

GRMHD simulations of equal-mass binary BHs (mass ratio $q = 1$, total mass $M = 2 \times 10^7 M_\odot$) in a circumbinary magnetized gas disk reveal that accretion is not steady but is modulated by the binary orbit. This produces periodic electromagnetic variability that serves as a potential pre-merger EM counterpart to gravitational waves detectable by LISA.

2. Accretion Rate Modulation

The total accretion rate modulation:

$$\dot{M}_{binary} \propto f_{orb} \cdot (1 + A \cdot \cos(\omega t + \phi))$$

where: - $f_{orb} \approx \frac{1}{2\pi} \sqrt{\frac{GM_{tot}}{r^3}}$ = binary orbital frequency - A = modulation amplitude (typically 0.1-0.5) - $\omega = 2\pi f_{orb}$ = angular orbital frequency - ϕ = phase offset

This modulation arises from the “lump” structure in the circumbinary disk that co-rotates at the beat frequency between the inner disk and binary.

3. Poynting Luminosity

The electromagnetic Poynting jet luminosity:

$$L_{Poynt} \propto \frac{B^2}{4\pi} \cdot v_A^2$$

where $v_A = B/\sqrt{4\pi\rho}$ is the Alfvén velocity. For magnetically arrested disk (MAD) conditions:

$$L_{Poynt,MAD} \approx \eta_{EM} \cdot \dot{M} \cdot c^2, \quad \eta_{EM} \approx 0.01$$

4. Orbital Frequency Term

Binary separation evolution due to GW radiation:

$$\frac{dr}{dt} = -\frac{64}{5} \cdot \frac{G^3 M_1 M_2 (M_1 + M_2)}{c^5 r^3}$$

$$f_{orb}(t) = \frac{1}{2\pi} \left(\frac{GM_{tot}}{r(t)^3} \right)^{1/2}$$

5. Spin Configurations

The accretion modulation depends on spin alignment: - **Aligned spins** ($a_\bullet = 0-0.9$): modulation amplitude $A \approx 0.3$ - **Anti-aligned spins**: kick velocities up to 3000 km/s, suppressed lump - **Precessing spins**: complex $\phi(t)$ time variation

For $a_\bullet = 0.9$, Lense-Thirring precession timescale:

$$\tau_{LT} = \frac{2\pi r^3}{2GJ/c^2}$$

6. Quadriadic UQFF Integration

The binary GRMHD accretion terms enter all four layers:

Layer 1 (bulk energy):

$$g_{L1,bin} = g_{UQFF} + \frac{\dot{M}_{binary} \cdot c^2}{r^2}$$

Layer 2 (resonance modulation):

$$g_{L2,bin} = L_{Poynt} \cdot a_\bullet + \frac{\dot{M}_{binary} \cdot \omega}{r}$$

Layer 3 (buoyancy/EM):

$$g_{L3,bin} = \frac{\dot{M} \cdot \omega \cdot \cos(\omega t + \phi)}{r^2}$$

Layer 4 (Q-wave/spin coupling):

$$g_{L4,bin} = \frac{L_{Poynt}}{r^2} \cdot a_\bullet$$

7. System Parameters (from arXiv:2309.03949)

- Binary mass: $M_{tot} = 2 \times 10^7 M_{\odot}$
 - Mass ratio: $q = 1$ (equal mass)
 - Spin: $a_{\bullet} = 0-0.9$
 - Gas configuration: prograde circumbinary disk
 - Simulation: GRMHD + magnetic field, ~ 5 orbital periods
 - Key result: $\dot{M}_{binary} \approx 0.045 \dot{M}_{Edd}$, modulated at f_{orb}
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8. LISA EM Counterpart Implications

Pre-merger LISA signal expected 1-10 years before coalescence for $M_{tot} \sim 10^7 M_{\odot}$ at $z < 1$. The orbital modulation in the accretion rate produces: - X-ray/optical variability at f_{orb} and $2f_{orb}$ - Quasi-periodic eruptions (QPEs) in soft X-ray band - Radio jets pulsing at orbital period

These are now trackable via the UQFF Layer 2 Resonance modulation term.

9. Summary

GRMHD binary BH simulations reveal accretion modulation $\propto (1 + A \cos(\omega t + \phi))$ proportional to orbital frequency, with Poynting-spin coupling $L_{Poynt} \cdot a_{\bullet}$. These terms formalize the GRMHD binary electromagnetic counterpart within the Quadriadic UQFF Layers 2-4.

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